

WHEN DOES A SPOKEN AGENT HAVE ENOUGH EVIDENCE TO ACT?

THE PACT-SLM CONTRACT TEST

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ABSTRACT

Streaming spoken agents may produce the correct final action after acting too early. Final-turn scores do not reveal whether each observed speech prefix supports an exposed action. We introduce the Partial Speech Action Contract for Turn Taking in Speech Language Models (PACT-SLM), a controlled test that assigns the first valid action time and evaluates both action identity and timing. In the primary test, 80 paired contrast groups from four held-out semantic families yield 1,600 prefix predictions across clean and 15 dB noise renderings. After correcting a branch-code/semantic-label mismatch, a refitted WavLM Base Plus probe reaches 26.03% pooled post-onset semantic-label accuracy (95% group-bootstrap interval [22.14%, 29.68%]), exposes an action on 18.99% of pre-onset prefixes, and predicts only 5.94% of complete trajectories exactly. Its pooled label score is at the 96th percentile of 100 within-prefix label permutations and below their 97.5th-percentile reference (26.73%). It exceeds matched text, scalar-acoustic, and shuffled-representation probes in pooled post-onset label accuracy, but the permutation result does not establish a robust speech-specific effect. Elapsed time is more onset-exact (36.25% versus 23.13%) while less accurate about action identity (9.92% versus 26.03%). These results show why action identity and timing need separate measurements; they do not establish a reliable streaming policy or generalization beyond this generated-speech diagnostic.

1 INTRODUCTION

Spoken agents must often decide before a user finishes speaking. They may continue listening, reason internally, ask for clarification, call a tool, or interrupt. A final-response score can verify the last action and still miss the central online error: the system acted before the speech available at that moment supported the action. That distinction matters whenever an exposed action changes external state or interrupts the user.

Current speech language models (SLMs) and full-duplex systems make early action possible, but model capability does not define a valid timing test (Chiang et al., 2026a;b; Défossez et al., 2024; Ge et al., 2025; Lin et al., 2025). The evaluator must know which prefix was visible, when an action first became valid, and whether a model used action-bearing evidence or a shortcut such as elapsed duration. Without this contract, high final accuracy can reward a system that guesses the action late, acts early, or simply recognizes an endpoint.

We ask: *what must an evaluation control to determine whether partial speech contains enough evidence for action?* PACT-SLM answers with a paired construction.¹ Two generated utterances share a transcript and observed audio through the last prefix labeled WAIT, then diverge before the first prefix at which their different actions become valid. All later prefixes are nested crops of one branch waveform. Training uses clean speech; testing holds out both semantic families and a 15 dB noise rendering. Matched controls expose duration, scalar-acoustic, text, and representation shortcuts.

¹Code and public release materials: <https://github.com/MENGZHEGENG/pact-slm>.

The corrected replay changes the interpretation. The resulting test separates action identity from action timing, but its action-identity evidence is limited: WavLM Base Plus exceeds matched text, scalar-acoustic, and shuffled-representation controls after onset, while its observed score falls below the 97.5th-percentile reference in a 100-permutation analysis. It also exposes actions before onset and predicts few complete trajectories. Elapsed time identifies the onset more often while identifying the action less often. A single final-action score cannot express this tradeoff.

Contributions.

1. We define a variable-onset evaluation contract that keeps paired pre-onset observations identical and preserves nested prefixes when noise is added.
2. We instantiate the contract as a 340-group generated-speech diagnostic with five prefixes per trajectory and matched controls for endpoint, elapsed time, authored text, scalar acoustics, and shuffled speech representations.
3. We reanalyze stored embeddings from 80 held-out groups using source-index semantic targets after finding that randomized branch codes do not have a fixed action meaning. The corrected probe shows limited pooled post-onset semantic-label accuracy, below the 97.5th-percentile reference of a 100-permutation check, alongside early exposure and low complete-trajectory accuracy. This supports separate measurement of identity and timing, not a robust speech-specific or deployable-policy claim.

2 RELATED WORK AND EVALUATION GAP

Speech encoders and modality adapters provide input interfaces for SLMs, including WavLLM, COSMIC, and AlignFormer (Hu et al., 2024; Pan et al., 2024; Fan et al., 2024). Simultaneous and full-duplex models address listening, reasoning, speaking, and interruption during an ongoing turn (Chiang et al., 2026a;b; Défossez et al., 2024). Broader duplex benchmarks evaluate conversation quality and turn management (Ge et al., 2025; Lin et al., 2025). PACT-SLM complements these systems by testing the evidence boundary for one decision: whether the current user-speech prefix supports an exposed action.

Prosody and contextual speech benchmarks evaluate paralinguistic or semantic reasoning (Qian et al., 2025; Wang et al., 2025; 2026). Their examples can reveal whether a representation encodes acoustic information, but an action-timing evaluation also needs a known onset, strictly nested observations, and controls that separate action evidence from time. We use frozen WavLM representations (Chen et al., 2022) as a reproducible probe, not as a new SLM architecture.

3 PACT-SLM EVALUATION CONTRACT

3.1 WHAT THE TEST MUST IDENTIFY

Let trajectory i have ordered speech prefixes $p_{i,1}, \dots, p_{i,K}$, target action a_i , and first valid action index k_i^* . The target at prefix k is

$$y_{i,k} = \begin{cases} \text{WAIT}, & k < k_i^*, \\ a_i, & k \geq k_i^*. \end{cases} \quad (1)$$

The evaluation must therefore answer two questions. Does the model identify a_i after the relevant evidence appears? Does it keep waiting before k_i^* and change state at the correct prefix? Final-action accuracy answers only the first question at $k = K$.

3.2 PAIRED BRANCH CONSTRUCTION

Figure 1 contrasts a fixed-endpoint shortcut with the PACT-SLM test. Each contrast group contains two transcript-matched source utterances with different actions. We assign k_i^* to prefix fraction 0.4, 0.6, or 0.8, balanced within split and action-pair buckets. The source waveforms begin to diverge at the temporal midpoint between the endpoint of the preceding observed prefix and the endpoint of the first action-labeled prefix. Before that frame, the two branch waveforms use the same averaged

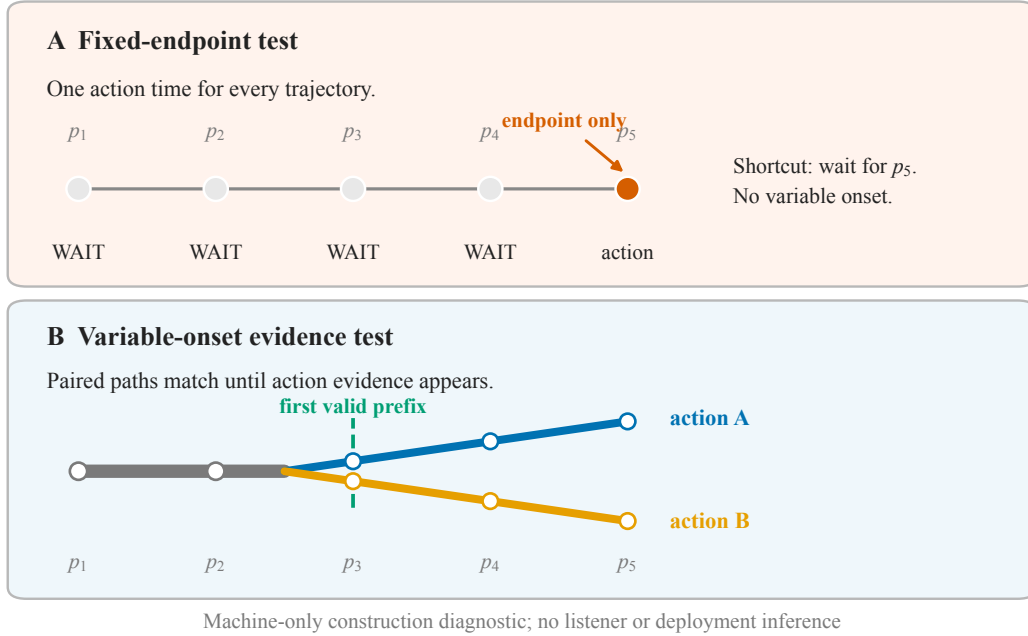


Figure 1: The PACT-SLM contract test. In the fixed-endpoint test (top), an action is valid only at the final prefix, so a duration or endpoint rule can wait until the last crop. In the variable-onset test (bottom), paired observed prefixes remain identical through the last required wait, action-specific evidence appears before the first valid action prefix, and every clean or noisy prefix is cropped from one branch waveform. Held-out families and noise test whether a probe uses transferable action evidence.

samples. A 50 ms crossfade then connects each branch to its own source waveform. The observed prefixes before onset are byte-identical within a pair, while the first action-labeled prefix contains branch-specific samples. This first-valid index is assigned by the construction; it is not a listener judgment of when an action becomes appropriate.

For each branch, prefixes at fractions $\{0.2, 0.4, 0.6, 0.8, 1.0\}$ are cropped from one complete waveform. The noisy rendering adds one shared pair-level Gaussian realization at 15 dB signal-to-noise ratio to each complete branch before cropping. This order preserves nesting and prevents independently sampled prefix noise from acting as an identifier. Here, the construction is a controlled test of the information contract; it is not a perceptual model of natural turn evolution.

3.3 DATA AND SPLIT

The initial generated-speech corpus contains transcript-matched action contrasts. An availability audit found 120 complete pairs whose referenced audio was absent; these pairs were excluded before fitting or scoring. Of the 360 audio-backed pairs, 340 have two distinct target actions and enter the diagnostic. The remaining 20 same-action pairs are ineligible for the paired branch test. Here a *contrast group* is one eligible pair of source utterances, a *source trajectory* is one utterance in that pair, and a *prefix record* is one of five crops of a source trajectory in one rendering. Thus, 208 training groups yield 2,080 clean prefix records; 52 development groups yield 520 clean records; and 80 test groups yield 1,600 records because both branches have five clean and five noisy crops. The total is 4,200 records from 680 distinct source trajectories, not 4,200 independent examples. Table 1 reports these units separately. The 80 contrast groups, each retaining both branches and renderings during resampling, are the bootstrap units.

The test split holds out four semantic families: book uncertainty, calendar noise, message barge-in, and timer urgency. Their 80 contrast groups do not occur in training or development. Clean speech is the only training and development rendering. Test includes both clean speech and the held-out

Table 1: PACT-SLM diagnostic-set composition after the audio-availability and action-contrast gates. A group is a paired action contrast; a source trajectory is one utterance in that group; a prefix record is one scored crop in one rendering. The 13 family entries in each development split are memberships, not additive unique-family counts. Test includes clean and held-out 15 dB noise renderings.

Split	Family memberships	Groups	Source trajectories	Prefix records	Rendering
Train	13	208	416	2,080	Clean
Development	13	52	104	520	Clean
Test	4	80	160	1,600	Clean, 15 dB noise

Table 2: Corrected variable-onset results on 80 held-out contrast groups across clean and 15 dB noise renderings (1,600 prefix predictions). Values are percentages. Trajectory exactness requires all five decisions for one trajectory and rendering to match. Exposed-action rate is calculated over pre-onset prefixes only and is lower when better; pooled post-onset semantic-label accuracy and onset exactness are higher when better. The highlighted entry is WavLM Base Plus refit on stored training embeddings with source-index semantic targets. Intervals are in Appendix D.

Model or control	Trajectory exact (%) $\uparrow$	Pre-onset exposed (%) $\downarrow$	Post-onset label (%) $\uparrow$	Onset exact (%) $\uparrow$
Always wait	0.00	0.00	0.00	0.00
Endpoint plurality	0.00	0.00	4.13	0.00
Elapsed time only	0.00	4.43	9.92	36.25
Authored text only	3.75	58.23	18.60	16.25
Scalar acoustics	1.56	10.13	14.88	18.13
Prefix-shuffled WavLM	0.31	51.58	15.29	17.81
WavLM Base Plus	5.94	18.99	26.03	23.13
WavLM Base Plus + authored text	6.56	15.19	25.10	23.13

noise rendering. Table 1 lists 13 family memberships in each of the training and development splits; those membership counts are not additive, and their overlap is not established by the post-audit result files. This distinction matters because a count of split memberships is not a count of unique families. The test design combines family and renderer transfer, with the clean slice preserving a matched reference.

4 MODELS AND MEASUREMENTS

Frozen speech representation. We extract the final hidden state from the TorchAudio WAVLM_BASE_PLUS checkpoint for every prefix and mean-pool it over time. The resulting 768-dimensional vector is standardized using training data. The audio-only probe is balanced multinomial logistic regression with $C = 1$, the `lbfgs` solver, random state 0, and a 2,000-iteration limit. No model or threshold is selected on the test set. The audio-plus-text probe concatenates the standardized speech vector with term frequency-inverse document frequency (TF-IDF) word and bigram features.

Matched controls. Always-wait measures the cost of never acting. Endpoint plurality receives an oracle final-prefix indicator and emits the most frequent training action only at that endpoint. Elapsed-only uses prefix duration. Text-only uses an authored token-fraction transcript, which is a controlled text feature and not a time-aligned transcript. Scalar-audio uses duration, root-mean-square energy, mean absolute amplitude, zero-crossing rate, and crest factor. For each of the five prefix positions, prefix-shuffled WavLM permutes the correspondence between training speech embeddings and action labels across examples with seed 0. This preserves position-specific action frequencies while breaking the speech-action association. Every learned probe uses the same classifier and training rows.

Table 3: Corrected WavLM Base Plus results by held-out rendering. Values are percentages with 1,000 contrast-group-bootstrap 95% intervals.

Metric	Clean	15 dB noise
Trajectory exact $\uparrow$	11.25 [6.25, 17.50]	0.62 [0.00, 1.88]
Pre-onset exposed $\downarrow$	21.52 [14.57, 29.17]	16.46 [10.69, 23.32]
Pooled post-onset label $\uparrow$	38.43 [31.76, 45.38]	13.64 [9.80, 17.72]
Onset exact $\uparrow$	40.00 [29.38, 49.38]	6.25 [1.88, 11.88]

Measurements and uncertainty. Trajectory exactness requires all five actions for one source and rendering to match Equation 1. Pre-onset exposed-action rate counts ASK-CLARIFY, CALL-TOOL, or INTERRUPT-SAFE before k_i^* ; internal THINK and ABSTAIN are not counted as exposed actions. SPEAK is not among the six labels in this diagnostic. Pooled post-onset semantic-label accuracy evaluates the source-index label at and after k_i^* across both exposed-action and internal-state prefixes. Onset exactness checks whether the first non-WAIT prediction occurs at k_i^* . We report 1,000 group-bootstrap repeats over the 80 contrast groups and paired intervals for model differences.

Reproducibility. The appendix specifies exclusions, split counts, onset assignment, branch and noise construction, model settings, semantic target repair, scoring rules, and uncertainty calculations. The post-audit analysis refits probes on stored embeddings and replays their saved predictions; it is not the historical fit or an independent replication. The underlying generated audio and embeddings are not contained in the manuscript source.

5 RESULTS

5.1 ACTION IDENTITY AND ONSET ARE DIFFERENT CAPABILITIES

Table 2 gives the corrected full-test comparison. WavLM Base Plus reaches 26.03% pooled post-onset semantic-label accuracy [22.14%, 29.68%], predicts 5.94% of complete trajectories exactly [3.44%, 9.06%], and exposes an action on 18.99% of prefixes strictly before the assigned onset [14.19%, 24.85%]. The pooled post-onset score is $252/968$: $195/606 = 32.18\%$ on exposed-action labels and $57/362 = 15.75\%$ on internal-state labels. It therefore is not an exposed-action-only score. These intervals resample the 80 contrast groups while holding the refitted models and saved predictions fixed; they do not include fitting uncertainty. The early-exposure denominator contains only prefixes before onset, not all 1,600 test records. The result is evidence of imperfect action and timing behavior in this probe, not a reliable timing policy.

The matched controls qualify this result. WavLM Base Plus exceeds prefix-shuffled WavLM by an absolute difference of 10.74% in pooled post-onset label accuracy [6.73%, 14.56%], and scalar acoustics by 11.16% [5.66%, 15.95%]. However, in 100 within-prefix permutations of the training-label correspondence, the observed 26.03% score is at the 96th percentile and below the 97.5th-percentile value of 26.73%. The probe also exceeds the text-only control by 7.44% [2.62%, 12.60%], but this permutation check does not support a robust speech-specific effect at the stated reference. The corrected findings are therefore consistent with a limited action signal in this fixed generated-speech test; they do not isolate a general speech-representation effect.

Elapsed time exposes a separate tradeoff. It reaches 36.25% onset exactness, compared with 23.13% for WavLM, a WavLM-minus-elapsed absolute difference of -13.12% [-23.76%, -1.88%]. Elapsed-only post-onset accuracy is 9.92%, compared with 26.03% for WavLM, a difference of 16.12% [10.85%, 21.50%]. Thus, duration predicts the assigned onset more often but identifies the target action less often. Neither score alone describes the joint timing-and-action behavior.

Figure 2 displays estimates and group-bootstrap intervals for all four outcomes. WavLM Base Plus has higher pooled post-onset label accuracy than the matched learned controls, while elapsed time has higher onset exactness. Under held-out noise, WavLM’s pooled post-onset label accuracy falls from 38.43% [31.76%, 45.38%] on clean speech to 13.64% [9.80%, 17.72%]; complete-trajectory exactness falls from 11.25% [6.25%, 17.50%] to 0.62% [0.00%, 1.88%]. These clean/noise intervals resample groups, and the same groups contribute both renderings.

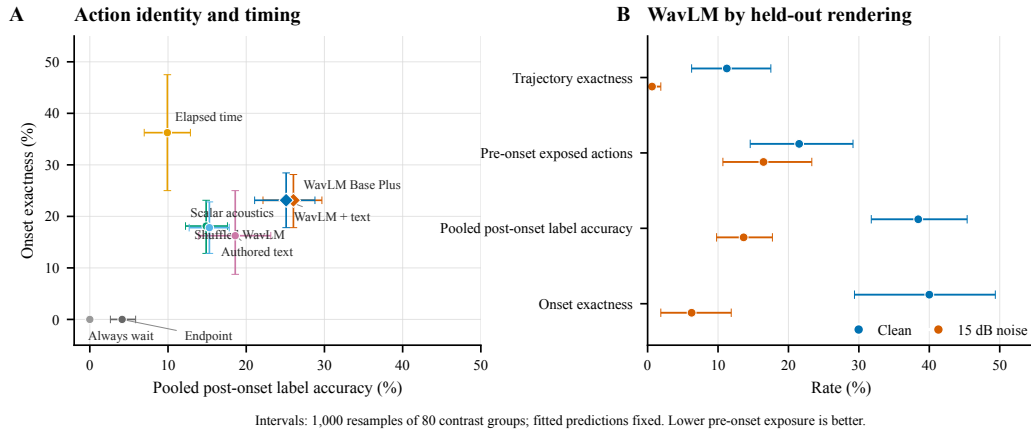


Figure 2: Corrected results on the 80-group held-out test. Each panel reports one metric for all scored probes; points and intervals are percentages and 95% intervals from 1,000 resamples of the 80 contrast groups. Intervals hold fitted probes and saved predictions fixed. The audio-only probe exceeds matched learned controls on pooled post-onset label accuracy, while elapsed time has higher onset exactness. Complete-trajectory accuracy remains low.

5.2 HELD-OUT NOISE BREAKS COMPLETE TRAJECTORIES

Table 3 shows the rendering split. WavLM trajectory exactness is 11.25% on clean speech and 0.62% under held-out 15 dB noise. Pooled post-onset label accuracy falls from 38.43% to 13.64%, and onset exactness falls from 40.00% to 6.25%. Adding authored text changes the combined pre-onset exposed-action rate from 18.99% to 15.19%, an absolute difference of -3.80% for the text-augmented probe minus the audio-only probe [-6.00%, -1.94%]. Its pooled post-onset label difference is -0.93% [-2.06%, 0.10%] and its trajectory-exactness difference is 0.63% [-0.31%, 1.88%]. In this comparison, text reduces early exposed actions but does not improve action identification; the tradeoff is not an overall gain.

5.3 WHY THE CONTRACT CHANGES THE CONCLUSION

As detailed in Appendix B, the earlier package and fixed-endpoint prototype illustrate why endpoint scores alone do not test online action timing. The package has one chunk per utterance and no WAIT target; a train-fitted condition lookup reaches 93.75% on both development and test. The first prototype adds prefixes but labels every action only at the final crop, allowing an endpoint-plurality diagnostic to avoid every early error. These artifacts can test final action recognition, but not whether the current prefix supports an action.

The variable-onset diagnostic instead evaluates a joint timing contract. Pair-identical wait prefixes prevent branch identity from leaking early, and source-index targets specify the semantic action at each first-valid prefix. The corrected analysis finds low complete-trajectory accuracy and an action score that does not exceed the permutation reference. This does not establish that speech carries no action information; it shows that the available post-audit evidence is insufficient for a robust speech-specific conclusion and that the tested probe does not solve the joint timing problem.

6 SCOPE AND LIMITATIONS

PACT-SLM is a controlled diagnostic construction, not a natural-speech benchmark or deployed policy evaluation. Its waveforms join an averaged shared prefix to one of two generated branches, and no listener study establishes naturalness, intelligibility, or human agreement with the action onset. Six training-development families were excluded because 120 complete source pairs lacked audio; another 20 audio-backed pairs had no action contrast. The held-out families had appeared in

earlier project development, so this test is exploratory evidence, not independent confirmation on an untouched corpus.

The evaluated model is a frozen WavLM Base Plus representation with a shallow classifier refit after target correction. The post-audit refit is not an independent replication, and historical fit lineage is not established. The authored text control is truncated by token fraction and is not a timestamped automatic transcript. The results concern this representation and construction. Native streaming SLMs, timestamped transcripts, natural speech, additional voices and synthesis systems, and human action-appropriateness judgments require separate evidence.

7 CONCLUSION

PACT-SLM makes an evaluation question explicit: recognizing an eventual action and timing its exposure are distinct measurements. In a corrected post-audit refit on stored embeddings, WavLM Base Plus reaches 26.03% pooled post-onset semantic-label accuracy, but its score is below the 97.5th-percentile reference from 100 within-prefix permutations. It exposes actions before onset and predicts only 5.94% of complete trajectories exactly; elapsed time is more onset-exact. The evidence supports using separate action and timing measures in this controlled generated-speech test. It does not establish a robust speech-specific effect, a reliable policy, or performance on natural conversation.

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Table 4: Reader-facing map of the three evaluation designs. Group counts, scored observations, and their nested units are separated explicitly. The earlier studies are design diagnostics, not additional test cohorts for the variable-onset results.

Design	Contrast groups	Scored observations and nested units	Question and inference limit
Earlier one-chunk package	Not reported in the paper source	960 utterance records total (384 train, 192 development, 384 test)	One 1.5-second chunk per utterance; there is no WAIT target, so action timing is not measured.
Fixed-endpoint prototype	64 total (40 train, 8 development, 16 test)	1,280 prefix records from 128 trajectories (five crops per trajectory, two renderings)	Every action is labeled only at the final crop. This tests endpoint recognition, not an intermediate valid action time.
Variable-onset diagnostic	340 total (208 train, 52 development, 80 test)	4,200 prefix records from 680 trajectories; the 80 test groups are resampled for uncertainty	Five nested crops per trajectory; training and development use clean speech, while the test includes clean and 15 dB noise renderings across four held-out semantic families. It measures action identity and timing separately.

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A APPENDIX ROADMAP AND STUDY KEY

This appendix gives the design map, target correction, full uncertainty results, and interpretation limits needed to reproduce and assess the main-text claims. The main comparison concerns one primary study: the variable-onset generated-speech diagnostic. Two earlier artifacts are included only to explain why endpoint evaluation is insufficient; they are not replications or additional test cohorts for the primary result.

Counts and units. The unit relationships are: one contrast group contains two source utterances (two source trajectories); each trajectory has five nested prefix crops; a rendering is applied to the full trajectory before cropping; and each prefix record is one crop under one rendering. In the variable-onset diagnostic, 208 training groups produce 416 trajectories and 2,080 clean prefix records; 52 development groups produce 104 trajectories and 520 clean records; 80 test groups produce 160 trajectories and 1,600 records (both clean and noise15 renderings). Thus, 4,200 records are repeated observations from 340 groups, not independent examples. The group is the unit resampled for the reported intervals. “13” in the train and development split ledger counts family memberships in each split, not 26 distinct families; the available post-audit summary does not establish the exact train/development family overlap, so we make no disjointness claim for those two splits. Four test families are held out from both.

B EARLIER PACKAGE AND FIXED-ENDPOINT AUDITS

The earlier package and fixed-endpoint prototype answer narrower questions than the main variable-onset test. They are retained because they show why a high final-action score or a late-only target cannot establish whether an action is justified at the current prefix. Table 5 separates single-chunk records from paired groups and prefix records; none of these earlier counts should be added to the primary test denominator.

The earlier v0.32 package contains 960 generated-speech records. Every split has only `chunk_id=0`, all intervals are $[0, 1.5]$ seconds, and no record has a WAIT target. A train-fitted condition lookup reaches 93.75% on both development and test; condition plus intent reaches 100%. These manifest fields are metadata, not acoustic evidence. The package also has 42 train–development, 126 train–test, and 42 development–test template overlaps. Because the test split repeats templates, these lookup scores do not show recognition of action-bearing speech.

The fixed-endpoint prototype contains 64 contrast groups: 40 training, 8 development, and 16 test groups. Each group contributes two source trajectories, five crops per trajectory, and two renderings, for 1,280 prefix records. Every action begins at the final crop, so an endpoint rule is told when it may act. Noise is sampled independently after cropping, and the authored transcript is truncated by token fraction. The endpoint-plurality diagnostic reaches 37.50% final-action accuracy with no early non-WAIT prediction. WavLM reaches 39.06% final-action accuracy but predicts a non-WAIT label on 25.78% of pre-final records. Every evaluated method has zero exact paired bundles. This prototype demonstrates the mismatch between final action and early exposure, but its construction cannot test intermediate valid onsets. These earlier values are retained only as design diagnostics and are excluded from the primary comparison.

Table 5: Earlier evaluation artifacts. Lookup values are fit on training records and evaluated on development or test records. The final column states what the artifact can establish.

Artifact or check	Train	Dev.	Test	Interpretation
Earlier one-chunk records	384	192	384	Final label only; no wait target
Condition lookup accuracy	–	93.75%	93.75%	Metadata lookup, not acoustic evidence
Condition plus intent lookup	–	100.00%	100.00%	Uses split-repeated template information
Fixed-endpoint groups	40	8	16	Action valid only at final crop
Fixed-endpoint prefix records	800	160	320	Repeated crops from 64 groups

C SEMANTIC TARGET CORRECTION AND REPLAY

The primary result was reanalyzed after a label-identity failure was found in the historical evaluator. The evaluator had interpreted `BRANCH_A` and `BRANCH_B` as if they named the same semantic actions throughout the dataset. They do not: the branch codes are counterbalanced, and their mapping to the source action changes across contrast groups. A branch identifier therefore cannot serve as an action target. A probe can learn an accidental branch-code convention and produce scores that are not semantic action accuracy.

The correction assigns WAIT to every prefix strictly before the assigned valid-onset fraction. At and after that onset, the target is the semantic action recorded for the exact source utterance in the source index. The replay checked 680 unique source utterance IDs and verified each manifest trajectory’s source action and counterbalanced branch code against its matching source-index entry. Probes were refit on the stored training embeddings under these corrected semantic targets; saved predictions were then scored against the corrected target. This is a post-audit refit and replay of stored embeddings. It is not the historical fitted model, a new data collection, or an independent replication. The available receipt does not establish the producer lineage of the historical fit.

The source-action oracle is included only as a label-wiring diagnostic. It receives the verified target action and onset by construction and therefore scores 100% on semantic action and trajectory metrics. It is not a model, deployable baseline, or evidence that acoustic information can recover the target. Its role is analogous to checking that the evaluator’s labels and metric implementation agree.

The corrected estimates can be recomputed from the saved prediction records with the evaluation procedure described above. Those records preserve the test utterance, contrast-group, trajectory, rendering, prefix, target-action, and predicted-action identities needed to rescore the reported metrics. The original generated speech and stored embeddings are not part of this paper source, so the package supports prediction-level rescoring but not independent reproduction of embedding extraction or probe fitting.

D FULL HELD-OUT RESULTS AND UNCERTAINTY

Table 6 reports the corrected estimate and 95% group-bootstrap interval for each model and each outcome. The inference unit is the contrast group: each resampled draw retains the two paired source trajectories, all available prefixes, and both test renderings. There are 80 groups and 1,000 bootstrap draws, with random seed 0. The fitted probes and their saved predictions remain fixed during resampling. These intervals quantify variation across the observed test groups conditional on those predictions; they do not include uncertainty from fitting, new generated speech, a new synthesis system, or natural conversations.

The important pattern is not a uniformly strong audio result. The corrected WavLM probe has 26.03% post-onset accuracy [22.14%, 29.68%], but exactness across all five decisions of a trajectory is 5.94% [3.44%, 9.06%]. Before onset, it exposes an action on 18.99% of eligible prefixes [14.19%, 24.85%]. Elapsed time is less accurate about the action (9.92% [6.96%, 12.86%]) but more often identifies the assigned onset (36.25% [25.00%, 47.50%]). Hence, the best action-identity score and best onset score belong to different controls, and neither approximates a reliable joint policy.

Table 6: Corrected estimates in percent with 1,000 group-bootstrap 95% intervals. Each cell is estimate [lower, upper]. The pre-onset exposure denominator contains only prefixes strictly before the valid onset.

Model or control	Trajectory exact $\uparrow$	Pre-onset exposed $\downarrow$	Post-onset label $\uparrow$	Onset exact $\uparrow$
Always wait	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]
Endpoint plurality	0.00 [0.00, 0.00]	0.00 [0.00, 0.00]	4.13 [2.63, 5.86]	0.00 [0.00, 0.00]
Elapsed time only	0.00 [0.00, 0.00]	4.43 [1.79, 7.69]	9.92 [6.96, 12.86]	36.25 [25.00, 47.50]
Text only	3.75 [1.25, 6.88]	58.23 [48.66, 68.02]	18.60 [13.93, 23.14]	16.25 [8.75, 25.00]
Scalar acoustics	1.56 [0.31, 3.12]	10.13 [7.10, 13.81]	14.88 [12.24, 17.60]	18.13 [12.81, 23.13]
Prefix-shuffled WavLM	0.31 [0.00, 0.94]	51.58 [46.73, 56.44]	15.29 [12.70, 17.83]	17.81 [12.81, 22.81]
WavLM Base Plus	5.94 [3.44, 9.06]	18.99 [14.19, 24.85]	26.03 [22.14, 29.68]	23.13 [17.81, 28.13]
WavLM Base Plus + authored text	6.56 [3.75, 10.00]	15.19 [10.60, 20.68]	25.10 [21.07, 28.79]	23.13 [17.81, 28.44]

E PAIRED DIFFERENCES AND COMPETING METRICS

Table 7 gives WavLM Base Plus minus each reference on the same 80 resampled groups. Differences are absolute differences in the named percentage-valued metric, computed from the paired group resamples; positive exposed-action differences mean more premature external actions and are unfavorable. Against prefix-shuffled WavLM, pooled post-onset label accuracy is higher by 10.74% [6.73%, 14.56%]. Against scalar acoustics, it is higher by 11.16% [5.66%, 15.95%]. These paired intervals quantify differences conditional on the fixed predictions; the separate permutation analysis below asks whether the score is unusual under altered training-label correspondence.

Elapsed time reverses the ranking for onset exactness: WavLM minus elapsed time is -13.12% [-23.76%, -1.88%]. This contrast illustrates why it would be misleading to say that WavLM is simply better overall. Text augmentation also reverses some metrics. Relative to audio only, audio plus authored text lowers pre-onset exposure by 3.80% [-6.00%, -1.94%], but its pooled post-onset label difference is -0.93% [-2.06%, 0.10%] and its trajectory-exactness difference is 0.63% [-0.31%, 1.88%]. Text changes the early-exposure outcome without a supported improvement in action identification.

Table 7: Paired WavLM Base Plus minus named reference differences in percentage-valued metrics, with 95% group-bootstrap intervals. Positive values favor WavLM for the upward metrics; a positive pre-onset exposure difference is unfavorable.

Reference	Δ trajectory exact	Δ pre-onset exposed	Δ post-onset label	Δ onset exact
Always wait	+5.94 [+3.44, +9.06]	+18.99 [+14.19, +24.85]	+26.03 [+22.14, +29.68]	+23.13 [+17.81, +28.13]
Endpoint plurality	+5.94 [+3.44, +9.06]	+18.99 [+14.19, +24.85]	+21.90 [+19.02, +24.90]	+23.13 [+17.81, +28.13]
Elapsed time	+5.94 [+3.44, +9.06]	+14.56 [+8.75, +21.39]	+16.12 [+10.85, +21.50]	-13.12 [-23.76, -1.88]
Text only	+2.19 [-1.56, +5.94]	-39.24 [-49.68, -28.02]	+7.44 [+2.62, +12.60]	+6.88 [-2.82, +15.31]
Scalar acoustics	+4.38 [+1.25, +8.12]	+8.86 [+2.90, +15.50]	+11.16 [+5.66, +15.95]	+5.00 [-0.94, +11.56]
Shuffled WavLM	+5.62 [+2.81, +8.75]	-32.59 [-38.26, -26.77]	+10.74 [+6.73, +14.56]	+5.31 [-0.31, +10.94]
Audio + text	+0.63 [-0.31, +1.88]	-3.80 [-6.00, -1.94]	-0.93 [-2.06, +0.10]	0.00 [-2.19, +2.19]

Permutation check. The corrected audio-only post-onset score is 26.03%. In 100 permutations of the training-label correspondence within prefix position (seeds 0–99), its empirical percentile is 96; the permutation median is 18.18%, and the 97.5th-percentile reference is 26.73%. Thus this score does not exceed that reference. This is a finite permutation diagnostic on one fixed split and one probe pipeline, not a calibrated population-level significance test. It constrains the interpretation: the observed score may reflect some structure captured by the probe, but this check does not justify a robust speech-specific claim. The permutation score is distinct from the 1,000 group-bootstrap intervals, which keep each model’s predictions fixed.

F RESULTS BY HELD-OUT FAMILY

The pooled test score conceals variation across the four held-out families. Table 8 reports descriptive point estimates for the audio-only and audio-plus-text probes. Each family contains 20 contrast groups, but the table’s cell-wise confidence intervals are not a correction for inspecting multiple families. Pooled post-onset label accuracy is highest for message barge-in (47.08% for audio only), while calendar noise is lowest (8.75%). Exact full trajectories remain uncommon in every family: 0.00% to 15.00% for audio only. This variability means the pooled estimate should not be read as uniform transfer across semantic families.

Table 8: Descriptive results within each of the four held-out semantic families (20 contrast groups per family). Values are percentages; these point estimates do not establish a between-family effect.

Family	Audio post-onset label	Text+audio post-onset label	Audio exposed before onset	Audio onset exact	Audio trajectory exact	Text+audio trajectory exact
Book uncertainty	19.67	17.62	14.10	21.25	0.00	1.25
Calendar noise	8.75	7.92	22.50	18.75	1.25	1.25
Message barge-in	47.08	47.50	12.50	22.50	15.00	16.25
Timer urgency	28.69	27.46	26.92	30.00	7.50	7.50

G CONSTRUCTION AND IMPLEMENTATION DETAILS

Availability and eligibility. The builder checks every referenced waveform and excludes a pair before onset assignment, fitting, or scoring if either source is missing. This removes 120 pairs from six training–development family memberships; all four held-out test families remain complete. Another 20 audio-backed pairs have identical target actions and do not form a paired action contrast. The resulting set is 340 groups: 208 training, 52 development, and 80 test. The fact that train and development each list 13 family memberships does not establish that those memberships are disjoint; that source-ID join is not available in the current summary.

Waveform and prefix construction. Each contrast group pairs transcript-matched source utterances with different actions. The assigned valid-onset fraction is 0.4, 0.6, or 0.8, balanced within split and action-pair buckets. The source branches diverge at the temporal midpoint between the end of the preceding observed prefix and the end of the first action-labeled prefix. Before divergence, branch waveforms use the same averaged samples; a 50 ms crossfade connects each shared segment

to its branch source. Prefixes are cropped at fractions $\{0.2, 0.4, 0.6, 0.8, 1.0\}$. For the noise condition, one pair-level Gaussian realization at 15 dB signal-to-noise ratio is added to each complete branch before cropping. This preserves nested noisy prefixes. The construction tests a controlled information contract, not perceived naturalness or listener agreement.

Probe and controls. The speech representation is the final hidden state from TorchAudio’s WAVLM_BASE.PLUS checkpoint, mean-pooled over time into a 768-dimensional vector and standardized using training data. The classifier is balanced multinomial logistic regression with $C = 1$, the `lbfgs` solver, random state 0, and a 2,000-iteration limit. Training and development use clean rendering only, and no threshold or model choice uses test scores. The audio-plus-text probe concatenates these standardized embeddings with word and bigram term frequency-inverse document frequency features. The text-only comparator uses authored token-fraction text rather than a time-aligned transcript. Scalar acoustics include duration, root-mean-square energy, mean absolute amplitude, zero-crossing rate, and crest factor. Prefix-shuffled WavLM permutes the pairing between training embeddings and action labels separately at each of the five prefix positions with seed 0; it retains position-specific action frequencies but removes the observed speech-to-action pairing. End-point plurality receives an oracle indicator of the final prefix and emits the most common training action only there. Always-wait never emits an action.

Metric definitions. For a trajectory, the target sequence is WAIT at every prefix strictly before its assigned onset and the source-index semantic action at and after onset. Trajectory exactness is the fraction of source-trajectory/rendering combinations for which all five predictions match. Pre-onset exposed-action rate is the number of predictions equal to ASK-CLARIFY, CALL-TOOL, or INTERRUPT-SAFE divided by the number of prefixes strictly before onset; internal THINK and ABSTAIN do not count as exposed actions. Pooled post-onset semantic-label accuracy is the number of correct source-index labels divided by all prefixes at or after onset. For WavLM, this pooled numerator and denominator are 252/968: 195/606 exposed-action labels (32.18%) and 57/362 internal-state labels (15.75%). Onset exactness is the fraction of source-trajectory/rendering combinations whose first non-WAIT prediction occurs at the assigned onset; trajectories that never emit a non-wait action are assigned a predicted onset of 1.2 for timing-error calculations and do not count as exact. SPEAK is not a label in this six-class diagnostic.

Uncertainty and interpretation. For each of the 1,000 percentile-bootstrap draws, 80 contrast groups are sampled with replacement. All records belonging to a sampled group are retained together so the two paired utterances and both test renderings remain associated; duplicated draws are treated as separate bootstrap copies. Paired model differences reuse the same sampled group IDs for both models. The seed is 0 and the intervals are percentile quantiles at 2.5% and 97.5%. This procedure describes test-group resampling conditional on the stored embeddings, fitted probes, and prediction file. It does not refit the probe in each draw. The 100-permutation distribution is a separate analysis and must not be conflated with bootstrap uncertainty.

Runtime and release limits. The recorded runtime uses PyTorch 2.6.0, TorchAudio 2.6.0, and CUDA 12.4; the embedding checkpoint has 94.4 million parameters and operates at 16 kHz. The supplementary MIT package contains the prediction-level replay script, contract tests, saved prediction records, and a public-safe reference report. It supports rescoring the saved predictions but does not regenerate speech, extract embeddings, or refit the probe. Raw generated speech and embeddings are not included here; the repository link is given in the Introduction.

H INTERPRETATION BOUNDARIES

This is a machine-only probe diagnostic on synthesized speech with assigned onset labels. It does not measure human judgments of action appropriateness or speech naturalness, or test a native streaming speech-language model that acts in conversation. The probe was refit after correcting semantic targets; the producer of the earlier fit remains unknown. The observed score did not exceed the 97.5th percentile of the 100-permutation reference distribution. These results characterize this probe, not general properties of speech representations or agent timing.